

Artificial Intelligence and Machine Learning Lab

Enrolment No: _____

Division: _____

Roll No: _____

Name: _____

Experiment No.: 04

Title: To study working of SVM algorithm and visualize the results.

Theory:

Support Vector Machine (SVM) is a powerful supervised learning algorithm used for classification and regression tasks. It is especially effective for high-dimensional data and binary classification problems.

Key Concepts:

1. Hyperplane:

In an SVM, a hyperplane is a decision boundary that best separates data points of different classes.

2. Support Vectors:

These are the data points that lie closest to the hyperplane. They **influence the position and orientation** of the hyperplane.

3. Margin:

SVM tries to maximize the **margin** — the distance between the hyperplane and the nearest data points from both classes.

4. Linearly Separable vs Non-linearly Separable Data:

Linear SVM works when data can be separated with a straight line (or hyperplane).

For non-linear data, **Kernel tricks** (like RBF, polynomial) are used to map the data into higher dimensions where it becomes linearly separable.

5. Kernel Functions:

Linear kernel: for linearly separable data

Polynomial kernel: maps features into polynomial space

RBF (Gaussian): suitable for complex and non-linear separation

6. SVM Parameters:

C: Regularization parameter; controls trade-off between correct classification and margin width

Gamma: Defines influence of a single training example in RBF kernel



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	Applications: • Text classification (spam detection) • Image classification • Face recognition • Bioinformatics (cancer classification) Visualization: In 2D datasets, SVM can be visualized by plotting: • The data points • The hyperplane • Support vectors (highlighted) • The margin boundaries
Pseudo code or Program	<pre> # Import required libraries import numpy as np import matplotlib.pyplot as plt import seaborn as sns from sklearn import datasets from sklearn.svm import SVC from sklearn.model_selection import train_test_split from sklearn.preprocessing import StandardScaler from sklearn.metrics import classification_report, confusion_matrix # Load the Iris dataset iris = datasets.load_iris() X = iris.data[:, :2] # Taking first two features for 2D visualization y = iris.target # Reduce dataset to only two classes for binary classification X = X[y != 2] y = y[y != 2] # Split the dataset X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42) # Feature scaling (very important for SVM) scaler = StandardScaler() X_train = scaler.fit_transform(X_train) X_test = scaler.transform(X_test) # Create and train SVM model model = SVC(kernel='linear', C=1.0) </pre>

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model.fit(X_train, y_train)

# Predict and evaluate
y_pred = model.predict(X_test)
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

# Function to plot decision boundary
def plot_decision_boundary(X, y, model, title):
    x_min, x_max = X[:, 0].min() - 1, X[:, 0].max() + 1
    y_min, y_max = X[:, 1].min() - 1, X[:, 1].max() + 1
    h = 0.02 # Mesh step size

    xx, yy = np.meshgrid(np.arange(x_min, x_max, h),
                          np.arange(y_min, y_max, h))

    Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
    Z = Z.reshape(xx.shape)

    plt.figure(figsize=(8,6))
    plt.contourf(xx, yy, Z, alpha=0.5, cmap=plt.cm.coolwarm)
    sns.scatterplot(x=X[:, 0], y=X[:, 1], hue=y, palette='coolwarm',
                    edgecolor='k')
    plt.title(title)
    plt.xlabel('Sepal Length (scaled)')
    plt.ylabel('Sepal Width (scaled)')
    plt.show()

# Visualize decision boundary
plot_decision_boundary(np.vstack((X_train, X_test)), np.hstack((y_train,
y_test)), model, "SVM Decision Boundary (Linear Kernel)")

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Outputs

(Students can attach extra pages if necessary)



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(Students can attach extra pages if necessary)

Exercise for Practice

Q1.	Explain the role of support vectors in SVM.
Ans:	
Q2.	What are advantages and disadvantages of SVM?
Ans	
Q3	What does the decision boundary plot tell you about the classifier's performance?
Ans	
Q4.	What is the main objective of the Support Vector Machine (SVM) algorithm?
Ans	



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Q5	In the Iris SVM lab, why do we select only two features and two classes for visualization?
Ans	
Conclusion	Thus, we have successfully Studied SVM Classifier and visualized it.

Date of Submission:	
Marks out of 10	Name and Sign of Subject Teacher
Remark:	

